

Project 1 Phase 1 – Problem Statement

UTAEngage – A Podcast Platform for UTA Students

UTA is planning to start “UTAEngage” - an online streaming platform exclusively dedicated to podcasts for UTA students, mirroring the concept of Spotify but focusing solely on podcast content. This platform serves as a comprehensive digital hub for podcast enthusiasts, offering a vast and diverse library of podcasts from a wide variety of themes.

The system, UTA is planning to design, will keep track of all Patrons, Artists, Podcasts, Episodes, Advertisers, Advertisements, as well as the relationships among them. This is an ideal solution for those seeking a dedicated platform that brings together several different podcasts, enabling access anytime and anywhere. This service is crafted to meet the evolving needs of modern audience, making it an essential tool for anyone who loves to listen to podcasts.

Data Description

Every UTAEngage user (that is a UTA student) has a unique netID and will have other basic information such as username, email, password, phone number, age, date of enrollment, cumulative GPA, enrolled department, nationality, and their full physical address comprising of the street address, apartment#, county and zipcode. These will be populated when they initially register on our platform. There are two types of users – patrons and artists. It is possible for a user to be both a patron and an artist.

Patron

- We define a patron as a UTA student who subscribes to podcasts, listens to daily/weekly episodes, follows/unfollows artists and engages with the episodes through likes and dislikes.
- The patrons will also have their listening related information stored to capture when, from which location and for how long they listened to a particular episode, and whether they listened to an episode multiple times.

Artist

- An artist is a student on our platform who creates and uploads new podcasts.
- Each artist has a unique artistID and an additional attribute that determines the number of followers they have. This number is different from the subscriber count of their individual podcasts.

Episode

- An episode is the audio/video file created and uploaded by artists on a particular podcast they own. Each episode has an episode number, which is stored in a sequentially increasing manner starting from 1.
- After a new episode is uploaded, all the podcast subscribers will get a notification.
- An episode will have attributes such as title, source - the URL where the actual file is physically stored, a format field which indicates whether it's an audio or video file, date

and time of upload, and the total duration. Additionally, important stats like the number of views, likes and dislikes for an episode can be calculated.

- An episode can only belong to a single podcast. When the podcast is deleted, all the episodes of that podcast are also deleted.

Podcast

- A podcast is defined as a set of episodes uploaded by artists, to which users can subscribe and get notifications when new ones are uploaded.
- An artist can create multiple podcasts and, of course, one podcast can have multiple episodes.
- Each podcast has a unique ID, name, date of creation, content description, number of subscribers and a pre-defined theme associated with it. Currently, supported themes are educational, travel, interview, storytelling, health and fitness, news, and technology.
- A podcast can only belong to a single theme. But an artist on the other hand can create multiple podcasts with different themes.

Advertiser

- An advertiser is a third-party company that is interested in advertising their ads on our platform.
- Each advertiser is identified by a unique ID, name, company headquarter city, and the number of employees.
- Advertisers are responsible for creating the ad content and providing our platform with a hyperlink.
- An advertiser can advertise multiple ads that may belong to different categories. If the advertiser is deleted, then all the ads belonging to that advertiser are also deleted.

Advertisement

- We define an ad as the promoted content of a registered advertiser that is displayed to patrons while they are listening to an episode of a particular podcast.
- It is characterized by the theme of the podcast which also matches the category of the advertisement created by the advertiser.
- Each ad has a unique ID, advertised product type, the hyperlink of the ad, the rate per view (paid to the advertiser by the platform), the predefined category it belongs to and the duration of the ad.
- Ads are not displayed on the podcast as a whole. In other words, an ad is associated with an episode that is displayed once at a certain timestamp.
- A single ad can appear in multiple episodes across different podcasts. For each episode where the advertiser decides to display an ad, a display cost needs to be paid to the platform. However, this cost can vary across episodes.

You are encouraged to **add more detail, if required, to the above data requirements** so that it helps you in the EER modeling phase. *Use red font color for the additions.*

Business Goals

1. For each day of the week, what is the average listening hours spent by a patron?
This is helpful for the artists to understand the patron activity and decide when to upload a new episode.
2. List the top 10 artists who on an average have received the most number of likes across all their podcast episodes uploaded in 2024.
Such a report helps the patrons to identify trending artists. Also, the ad display cost and rate per view can be adjusted for the episodes uploaded by such top artists.
3. For each nationality and year of birth, list the most popular theme along with its average episode duration. The popularity of a theme is quantified based on the total number of views the related podcast receives.
This report is helpful to gauge the listening pattern among different age groups belonging to diverse backgrounds. Based on such analysis, new patrons can be shown relevant content to keep them engaged and hooked to the platform.
4. List the name, theme, number of episodes, and the average number of views per episode for the podcasts that have only been listened to by the students from the 'CSE' department enrolled after July 2023. Order the list in the decreasing order of the average number of views per episode.
Such listicles help to identify trending podcasts among a certain group of people.
5. Who are the student subscribers who have listened to every video episode of the technology or educational themed podcasts that have the phrase 'large language model' appear somewhere in the description? Find their name, GPA, and email ID.
Such a report identifies the enthusiasts who are listening to a specific type of content. This set of people can act as a group who have acquired enough relevant knowledge on a certain topic and thus can become a part of related research, survey and promotional teams, or can be contacted for creating related podcasts.

In the above business goals, the advertisement information has not been used. You are required to think and clearly write down **two more interesting business goals which use the captured Advertiser and Advertisement information, along with other captured data.**

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